

Machine Learning-Based Healthy Meal Classification and Nutritional Pattern Analysis

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Course: Data Mining and Warehousing Lab

Abstract

Maintaining a healthy diet requires evaluating meals against multiple nutritional attributes such as calories, protein, carbohydrates, fat, sugar, and fiber, a task that is difficult to perform manually and consistently. This project presents a data mining and machine learning based framework for automatically classifying meals as *Healthy* or *Not Healthy* using the Healthy Meal Plan dataset obtained from Kaggle. The WEKA Java API is used to preprocess the data and to build, train, and evaluate three supervised classification algorithms: Random Forest, J48 Decision Tree, and Support Vector Machine (SMO), assessed through stratified 10-fold cross-validation using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and the Confusion Matrix. In addition to classification, K-Means clustering is applied to discover groups of nutritionally similar meals, and the Apriori algorithm is used to mine association rules that reveal frequent relationships among nutritional attributes. The best-performing classification model is serialized and reloaded to confirm that it can be reused for future predictions. The resulting system demonstrates a complete, data-driven pipeline from raw nutritional data to actionable dietary insight.

Keywords: Healthy Meal Classification, WEKA, Random Forest, J48, SMO, K-Means Clustering, Apriori, Association Rule Mining, Data Mining

1 Introduction

Diet plays a central role in maintaining human health, and the ability to quickly determine whether a given meal is nutritionally healthy is valuable for individuals, dietitians, and health-monitoring applications alike. Traditionally, this assessment is performed manually by comparing a meal's nutritional composition, such as calorie content, macronutrients, and sugar levels, against dietary guidelines. This process is subjective, time consuming, and difficult to scale across large numbers of meals or users.

With the growing availability of structured nutritional datasets, data mining and machine learning techniques offer an opportunity to automate this assessment. By learning patterns from labeled meal data, a classification model can generalize the criteria that distinguish healthy meals from unhealthy ones, while clustering and association rule mining can uncover hidden structures and relationships within the data that are not immediately obvious through manual inspection.

This project therefore proposes a complete data mining pipeline, implemented using the WEKA Java API, that combines supervised classification, unsupervised clustering, and association rule mining to analyze and predict the healthiness of meals based on their nutritional attributes.

2 Problem Statement

Manually determining whether a meal is healthy based on several interacting nutritional attributes, including calories,

protein, carbohydrates, fat, sugar, and fiber, is difficult and does not scale well to large datasets. There is a need for an automated, data-driven system that can (i) classify meals as *Healthy* or *Not Healthy*, (ii) group meals with similar nutritional profiles, and (iii) uncover frequent nutritional patterns associated with healthy or unhealthy meals.

3 Objectives

The objectives of this project are as follows:

1. Preprocess and transform the Healthy Meal Plan dataset into a suitable format for data mining using the WEKA Java API.
2. Develop a binary classification model to predict whether a meal is **Healthy** or **Not Healthy** based on its nutritional attributes.
3. Train and compare multiple supervised learning algorithms, including **Random Forest**, **J48**, and **SMO**, using stratified 10-fold cross-validation.
4. Evaluate the classification models using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and the Confusion Matrix.
5. Apply **K-Means** clustering to identify groups of meals with similar nutritional characteristics and analyze their relationship with the healthy/unhealthy classes.
6. Apply **Apriori** association rule mining to discover significant relationships among nutritional attributes and patterns associated with healthy meals.

7. Save the best-performing classification model and demonstrate its successful loading and reuse for prediction.

4 Methodology

4.1 Dataset Collection

The Healthy Meal Plan dataset is obtained from Kaggle and used as the primary source of data for this project. The dataset contains information describing different meals and their nutritional characteristics. Relevant attributes such as calorie content, protein, carbohydrates, fat, fiber, sugar, and other available nutritional features are used for analysis. A binary target attribute, **Healthy**, is used to label each meal as either *Healthy* or *Not Healthy* according to defined labeling criteria.

4.2 Data Preprocessing

The collected dataset is first converted from CSV format to ARFF format for compatibility with the WEKA Java API. Missing values are handled using the `ReplaceMissingValues` filter, and categorical attributes represented as strings are converted using the `StringToNominal` filter. Numerical attributes are normalized or standardized where appropriate so that attributes with different numerical scales do not disproportionately influence the learning algorithms. The dataset is then divided into input attributes and the target class attribute, with **Healthy** designated as the class attribute.

4.3 Exploratory Data Analysis

Basic statistical analysis is performed to examine the dataset's characteristics. Mean and standard deviation are considered for numerical attributes, and the distribution of the Healthy and Not Healthy classes is examined to identify potential class imbalance before model training.

4.4 Supervised Classification

Three classification algorithms are implemented using the WEKA Java API: **Random Forest**, **J48 Decision Tree**, and **Support Vector Machine (SMO)**. Each classifier is evaluated using stratified 10-fold cross-validation, and the models are compared using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and their confusion matrices. Where substantial class imbalance is observed, an appropriate technique such as SMOTE is considered to improve minority-class performance.

4.5 Unsupervised Learning

K-Means clustering is applied to the preprocessed nutritional attributes to identify groups of meals with similar nutritional

characteristics. The resulting clusters are analyzed to determine whether particular clusters contain a higher proportion of healthy or unhealthy meals, providing additional insight beyond the supervised labels.

4.6 Association Rule Mining

Selected numerical nutritional attributes are discretized into meaningful categories before applying the **Apriori** algorithm. Association rules are mined to discover relationships among nutritional characteristics and patterns associated with the Healthy and Not Healthy classes. The generated rules are evaluated using support, confidence, and lift, with emphasis placed on rules exhibiting high lift and meaningful nutritional relationships.

4.7 Model Persistence

The best-performing classification model, selected based on the evaluation results, is saved to disk using WEKA's model serialization functionality. The saved model is subsequently reloaded and used to demonstrate that a trained model can be reused for predicting the health category of new meals without retraining.

5 System Workflow

Figure 1 illustrates the overall workflow of the proposed system, from raw dataset acquisition through preprocessing, exploratory analysis, the three parallel mining tasks (classification, clustering, and association rule mining), and final model selection and persistence.

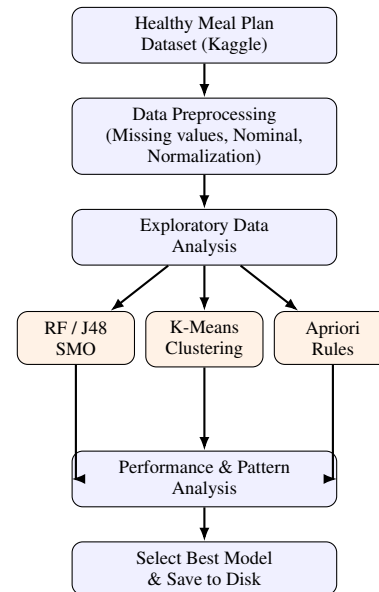


Figure 1: Overall workflow of the proposed data mining pipeline.

6 Tools and Technologies

- **Language:** Java
- **Data Mining Library:** WEKA Java API
- **Dataset Source:** Kaggle (Healthy Meal Plan Dataset)
- **File Formats:** CSV, ARFF
- **Algorithms:** Random Forest, J48, SMO, K-Means, Apriori

7 Evaluation Metrics

Model performance is assessed using standard classification metrics:

- **Accuracy** – overall proportion of correctly classified meals.
- **Precision** – proportion of predicted healthy meals that are truly healthy.
- **Recall** – proportion of actual healthy meals correctly identified.
- **F1-Score** – harmonic mean of precision and recall.
- **ROC-AUC** – ability of the model to distinguish between classes.
- **Confusion Matrix** – detailed breakdown of true/false positives and negatives.

8 Expected Outcomes

This project is expected to produce a reliable classification model capable of predicting whether a meal is healthy based on its nutritional characteristics. The performance of Random Forest, J48, and SMO will be compared using the metrics described above, and the best-performing model will be selected for final prediction. The clustering analysis is expected to reveal distinct groups of meals based on nutritional profile and to show whether specific clusters are predominantly associated with healthy or unhealthy meals. The Apriori analysis is expected to reveal meaningful associations among nutritional attributes, highlighting combinations of dietary characteristics frequently associated with healthy or unhealthy meals. Finally, the best-performing model will be serialized and successfully reloaded using the WEKA Java API, confirming a complete end-to-end pipeline from data preprocessing to model persistence and reuse.

9 Conclusion

This project outlines a comprehensive data mining and machine learning approach to healthy meal classification, combining supervised classification, unsupervised clustering, and association rule mining within a single WEKA-based pipeline. By systematically comparing multiple clas-

sification algorithms and complementing them with clustering and pattern-discovery techniques, the proposed system aims to deliver both an accurate predictive model and deeper insight into the nutritional factors that characterize healthy meals. The completed pipeline, including model persistence, demonstrates a practical and reusable framework applicable to real-world dietary analysis tools.

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